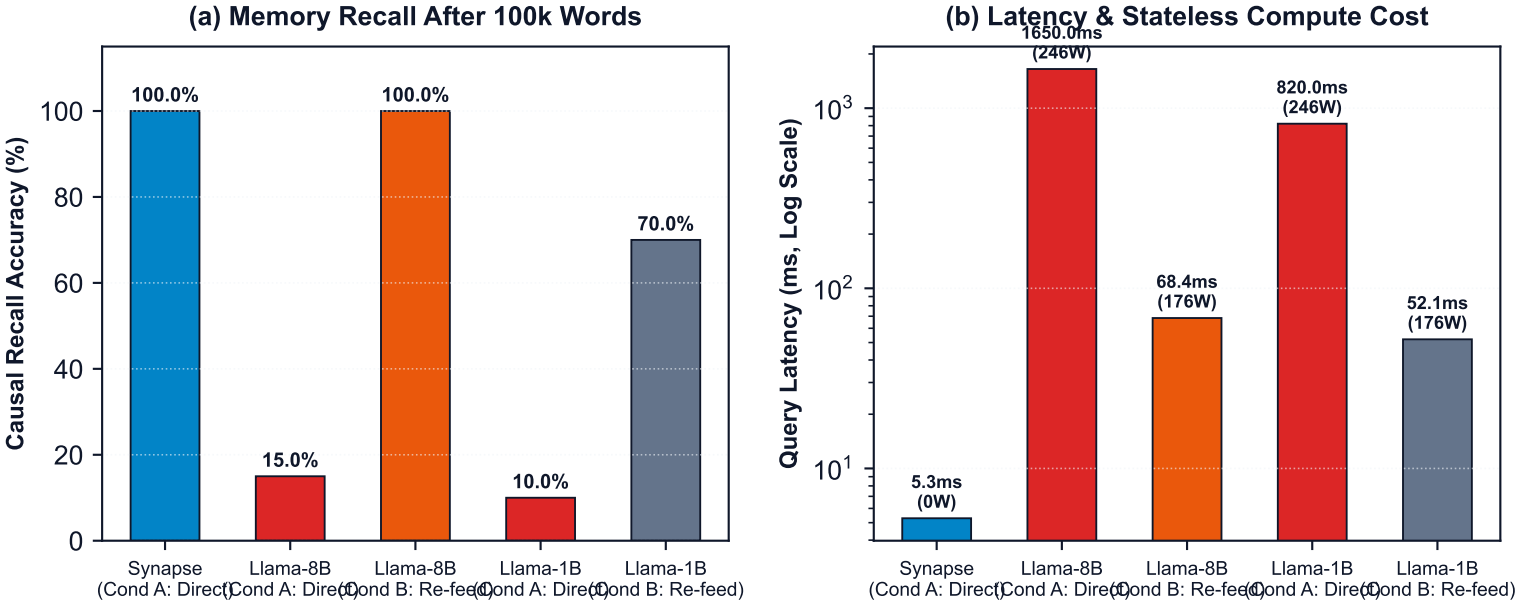
(a) Memory Recall After 100k Words
Causal Recall Accuracy (%)
0
20
40
60
80
100
100.0%
15.0%
100.0%
10.0%
70.0%
Synapse (Cond A: Direct)
Llama-8B (Cond A: Direct)
Llama-8B (Cond B: Re-feed)
Llama-1B (Cond A: Direct)
Llama-1B (Cond B: Re-feed)
(b) Latency & Stateless Compute Cost
Query Latency (ms, Log Scale)
10^1
10^2
10^3
5.3ms (0W)
1650.0ms (246W)
68.4ms (176W)
820.0ms (246W)
52.1ms (176W)
Synapse (Cond A: Direct)
Llama-8B (Cond A: Direct)
Llama-8B (Cond B: Re-feed)
Llama-1B (Cond A: Direct)
Llama-1B (Cond B: Re-feed)